



PHICOMP®



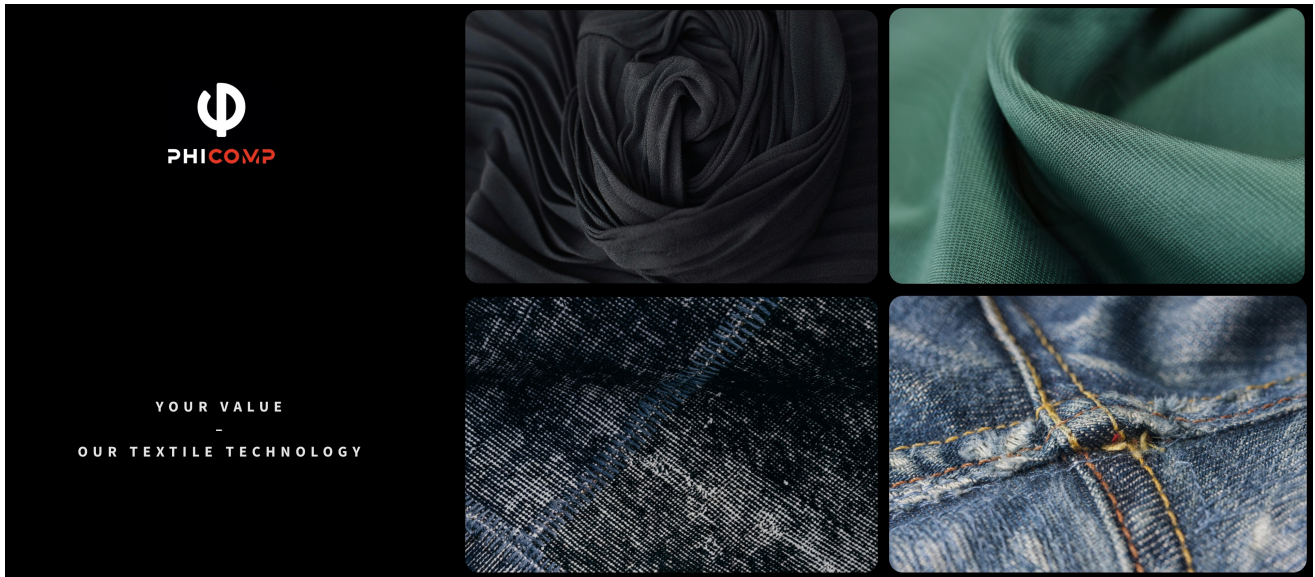
Product Portfolio

Category: Rotor Spinning

Product: Rotor

Status: August 2026





Your Value – Our Technology

- European coating quality rotor
- Consistent yarn quality and performance over lifetime
- Lowest power consumption per rotor cup
- Standard and high-speed rotor bearings
- Customized rotor design solutions
- Textile technology application advice and support

Rotor cup and bearing

Semi-automated rotor spinning machines



PHICOMP®



- **Coating** – High-End European nickel diamond coating
 - Superior coating for highest lifetimes
 - Consistent yarn quality and strength over lifetime
- **Curved thin wall design**
 - Reduced air friction decreases power consumption
 - Reduced moment of inertia enables quick start/stop
- **Bearing**
 - Standard bearings up to 100'000 rpm
 - High-speed bearing up to 110'000 rpm

Applicable for	• Schlafhorst/Oerlikon/Saurer BD3xx, BD4xx, BD6,7,8 • Rieter BT9xx, R3x • Taitan, Rifa and others
Rotor groove types	• Standard: T, V, Z, R, Tc U, Tr, and customized types
Rotor groove diameter	• 30mm to 46mm (>46mm **)
Material	• Steel: high-strength quality • Aluminum
Coating	• D(D) – Nickel diamond coating for extended lifetime • D(D)N – Nickel diamond coating + top (smooth) nickel layer • N – Smooth surface for sensitive application (steel rotor only)
Expected service life*	• D: > 20'000 h, DD: > 35'000 h
Warranty	• 12 months / 8000 h

* Specific application may lead to reduced lifetimes: sandy cotton, bleached cotton. man-made fibers with titanoxyd, viscose

		Order number				Order number	
Rotor type	Rotor speed	Cup	Rotor complete	Rotor type	Rotor speed	Cup	Rotor complete
C533/U-D	<110'000	10000074	10000075	T 32 D	<110'000	10000135	10000136
C536/U-D	<90'000	10000078	10000079	T 32 DN	<110'000	10000139	10000140
C536/U-DN	<90'000	10000220	10000221	T 33 D	<110'000	10000131	10000132
S536/U-D	<90'000	10000197	10000198	T 33 DN	<110'000	10000169	10000170
C338/U-D	<90'000	10000082	10000083	T 34 D	<110'000	10000002	10000003
C341/U-D	<90'000	10000086	10000087	T 34 DN	<110'000	10000125	10000126
C344/U-D	<70'000	10000090	10000091	T 34 DD	<110'000	10000147	10000148
C250/U-D**	<65'000	10000094	10000095	T 34 DDN	<110'000	10000150	10000151
C254/U-D**	<60'000	10000098	10000099	Ts 36 D	<90'000	10000012	10000013
C531/T-D	<110'000	10000062	10000063	Ts 36 DN	<90'000	10000006	10000007
C533/T-D	<110'000	10000066	10000067	Ts 40 D	<80'000	10000128	10000023
S533/Tr-D	<110'000	10000213	10000214	Ts 40 DN	<80'000	10000110	10000115
C533/Tr-D	<110'000	10000070	10000071	Ts 43 D	<75'000	10000026	10000027
C531/R-D	<110'000	10000106	10000107	Ts 43 DN	<75'000	10000113	10000114
C248/S-D**	<65'000	10000102	10000103	Vs 34 DN	<110'000	10000176	10000177
				Vs 36 DN	<90'000	10000122	10000123
				Vs 43 DN	<75'000	10000118	10000119
				Z 37 DN	<90'000	10000166	10000167
				Z 40 DN	<80'000	10000162	10000163
				Z 43 DN	<75'000	10000158	10000159
				Tc 36 DN	<90'000	10000193	10000194
				R 36 D	<90'000	10000217	10000218

** larger rotor diameters on request

Cannot find your rotor type or need a specific problem solution: please contact us



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01.08.2026
Version 1.6

Rotor cup and bearing

Semi-automated rotor spinning machines



• Rotor bearing

• High-speed rotor bearing

- **High productivity** due to rotor speeds up to 110'000 rpm
- **Long lifetime** due to superior materials applied (e.g.: special bearing steel and ceramic)
- **Options**
 - PC 105-9C1 – one relubrication port
 - PC 105-9C2 – two relubrication port

• Other rotor bearings

- <80'000 rpm : PC 80-10C1/2
- <90'000 rpm : PC 90-9C1/2
- <100'000 rpm : PC 100-9C1/2



• Diamond-Nickel Coating

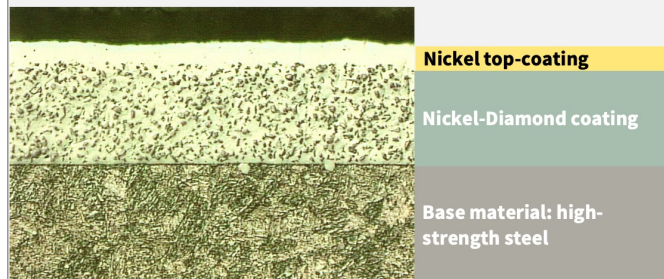
• European coating application

- **High yarn strengths and low CV** due to constant coating quality
- **High wear resistance (lifetime)** due to superior distribution of diamond in a hardened nickel basis
- **Low variation of technological yarn values over lifetime** due to superior evenness of the coating layer

• Options

1. DN-coating: **D + N**
2. D-coating: **D**
3. DD-coating: **D + D**
4. DDN-coating: **D + D + N**
5. N-coating: **N**

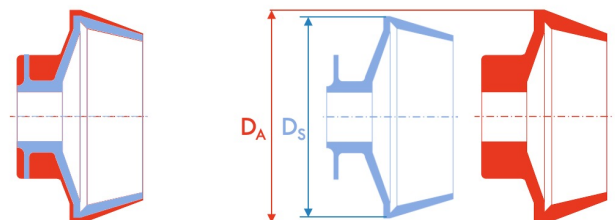
• Coating Design



• Advantage of Steel rotor cups

- **Lower power consumption** due to a smaller outer diameter of the steel rotor cup
 - Smaller Diameter = less air-friction = lower power consumption
- **Extended lifetime** due to higher strength and hardness of the steel
 - Rotor can be used for a limited time if the coating is already worn (replacement phase)
- **Lower sensitivity to spin finish** under coating surface corrosion (e.g., viscose, PES)

• Comparison



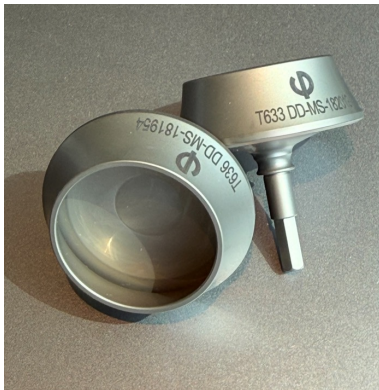
	Steel	Aluminum
Material strength	100%	appr. 50%
Material hardness	100%	appr. 40%
Weight of rotor cup	100 – 105%	100%
Outer diameter (D)	D _s	D _A =D _s +1.5mm



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- **Solid single-piece rotor** with excellent concentricity
 - Low noise and low power consumption
 - Reduced load on magnetic bearing and motor
- **Curved thin wall design**
 - Reduced air friction decreases power consumption
 - Reduced moment of inertia enables quick start/stop
- **Coating DD** – High-End European nickel diamond coating
 - Extended coating for highest lifetimes*
 - Consistent yarn quality and strength over lifetime

Applicable for	• Autocoro 8 to 11
Rotor groove types	• Standard: G, T, Ts, Tc(TT), K, U, S • PhiComp customized types: Tx
Rotor groove diameter	• 28mm to 46mm**
Coating	• DD – Nickel diamond coating for extended lifetime • DDN – Nickel diamond coating and top (smooth) nickel layer
Expected service life*	• DD: > 35'000 h
Warranty	• 12 months / 8000 h

* Specific application may lead to reduced lifetimes: sandy cotton, bleached cotton. man-made fibers with titanoxyd, viscose

Rotor type	Order number
G 628 DD	12000014
G 630 DD	12000017
G 631 DD	12000020
G 633 DD	12000023
G 636 DD	12000026
T 633 DD	12000002
Tx 633 DD	12000053
T 636 DD	12000005
T 640 DD	12000008
T 646 DD	12000011
Tc 636 DD	12000059
Tc 640 DD	12000050
Tc 646 DD	12000035
U 640 DD	12000044
S 652 DD**	12000041
K 631 DD	12000038

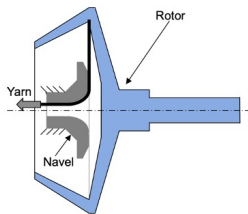
** larger rotor diameters on request

Cannot find your rotor type or need a specific problem solution: please contact us

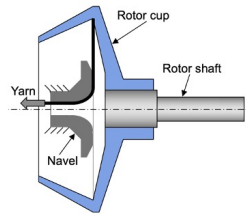


• Design SolidRotor

PhiComp



Others



• Single-piece design

1. Higher accuracy (run-out)
2. Less noise
3. Lower power consumption
4. Reduced load on Rotor Motor



• Diamond-Nickel Coating

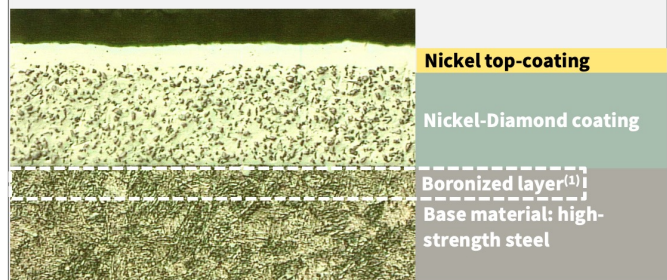
• European coating application

- **High yarn strengths and low CV** due to constant coating quality
- **High wear resistance (lifetime)** due to superior distribution of diamond in a hardened nickel basis
- **Low variation of technological yarn values over lifetime** due to superior evenness of the coating layer

• Options

1. DD-coating: **D + D**
2. DDN-coating: **D + D + N**

• Coating Design



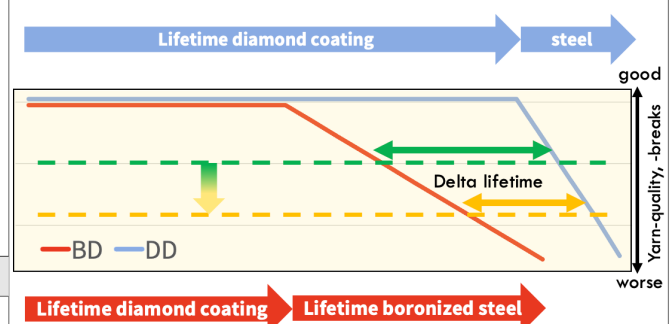
¹⁾ PhiComp rotors don't have (B) boronized layer underneath the nickel diamond coating

• Key facts BD vs. DD

- Diamond coating (D) defines the yarn quality
- The longer the spinning in diamond coating layer the longer the period of constant yarn quality
- Boronizing (B) is a thin layer hardening process of the steel, which is causing deformation, cracks and tension of the cup due to high process temperatures
- Yarn quality deteriorates gradually when spinning in the boronized steel or on the non-boronized steel

	BD	DD
Lifetime of rotor cup	100%	appr. 120%
Constant yarn quality period	100%	appr. 180%
Accuracy cup – run-out	worse	better
Power consumption due to run-out	higher	lower
Load on rotor motor due to run-out	higher	lower

• Yarn quality over lifetime



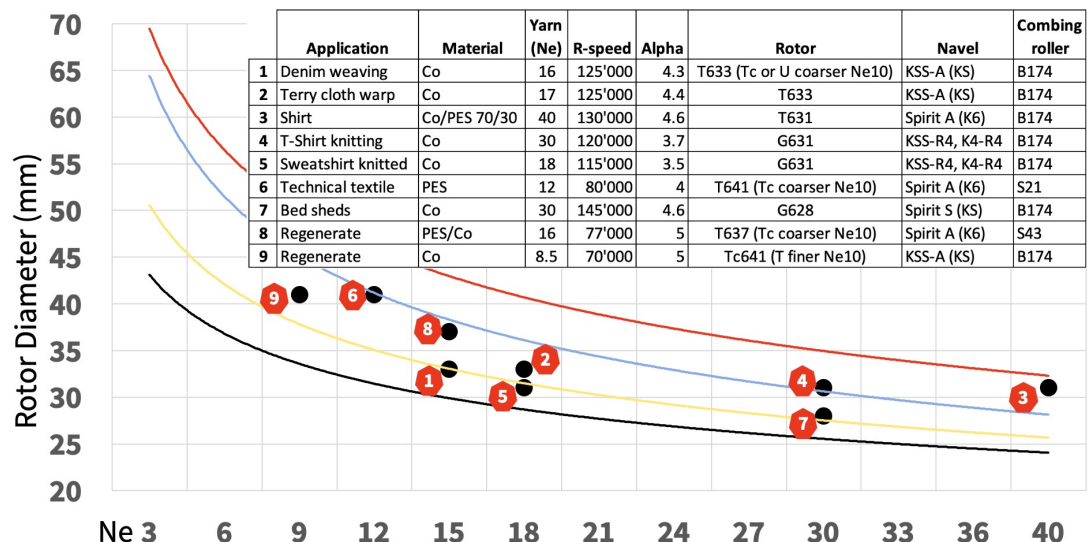
Applications and Technical data

Rotor speed and application range

- Ball bearing units are currently limited at appr. 110'000 rpm (semi-auto machinery)
- Smaller rotors (<26mm) at higher speeds are technically possible (fully automated)

	Yarn count (Ne)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21			22	23	24	27	28	29	30	31	32	33	34	35	36	37	38	39	40		
Speed range in rpm (ball rotor bearing limitation)	Rotor diameter (mm)																																										
140'000-160'000 (not applicable)	26																																										
125'000-150'000 (not applicable)	28																																										
95'000-135'000 (110'000)	30																																										
90'000-130'000 (110'000)	31																																										
85'000-125'000 (110'000)	33																																										
80'000-120'000 (110'000)	34																																										
70'000-110'000	36																																										
70'000-100'000	38																																										
65'000-90'000	40																																										
55'000-80'000	43																																										
45'000-75'000	46																																										
45'000-70'000	48																																										
40'000-65'000	50																																										
35'000-60'000	54																																										

Typical application (automated rotor machinery)



- **Rule of thumb: Rotor type**
 - **Knitting:** G-type rotor
 - **Weaving:** T-type rotor
 - **Coarser yarn counts:** Tc/U instead G or T
- **Rule of thumb: Navel type**
 - Good yarn quality → Smooth navel
 - High Spinning stability → navel with notches
 - High hairiness (bulkiness) → whirl inserts
 - Sensitive fibers → small funnel radius

	Co	Co/PES	Regenerate	Co	Co/PES	Regenerate
Finer Ne 10 Rotor speed > 110'000 rpm						
	K8 K4-R4 KSS-R4 KSK 6 Spirit A (K6)	KN-2R4 Spirit A (K6) K4	K4	KSS Spirit-S (KS) KSK 4 K4	Spirit-S (KS) K4	KSS
Coarser Ne 10 Rotor speed < 110'000 rpm				 		
	K8/36 K4	K8/36 K4	K8/36 K4	KN KSS K3 K4	K4	KN K4
	Knitting			Weaving		



Applications and Technical data

Fully-automated machine		Semi-automated machine	
Rotor groove shape	Application / Yarn characteristics	Rotor groove shape	Application / Yarn characteristics
	Ne 10 and finer - universally applicable - very good spinning stability (clean cotton) - good yarn values - higher hairiness, bulky yarn - lower yarn strength - risk of moirè faults - knitting yarns		Ne 10 and finer - universally applicable - very good spinning stability (clean cotton) - good yarn values - higher hairiness, bulky yarn - lower yarn strength - risk of moirè faults - knitting yarns
	Ne 10 and finer - universally applicable - good yarn values - low hairiness - high yarn strength - snarling tendency - less moirè faults - weaving yarn (most materials) - compact yarn		Ne 10 and finer - universally applicable - good yarn values - low hairiness - high yarn strength - snarling tendency - less moirè faults - weaving yarn (most materials) - compact yarn
	Application between T and Tc		Application between Tr and Tc
	Ne 10 and coarser, - good yarn values - low hairiness - high-bulky yarns - high yarn strength - snarling tendency - less moirè faults - weaving/knitting yarn (regenerate) - denim application		Ne 10 and coarser, - good yarn values - low hairiness - high-bulky yarns - high yarn strength - snarling tendency - less moirè faults - weaving/knitting yarn (regenerate) - denim application
	Ne 10 and coarser, - high-bulky yarns - higher hairiness - lower yarn strength than T or Tc - low snarling tendency - mostly for denim application		Ne 10 and coarser, - good resistance to fiber shifting - Man-made fiber - Secondary fibers
		(similar to V)	